

Volume mount

While using volumes, pay detailed attention, as this could be one of the most vulnerable options. Avoiding Docker socket mount is one of the best practices to secure Docker. Here is an example of volume mount that is extremely vulnerable for applications based on containers.

```
-v /var/run.docker.sock:/var/run/docker.sock
```

Securing the base machine

Check if the base/host machine is updated with the latest version of Docker and all the patches have been applied for the same.

Scanning image vulnerabilities

If you are pulling images from non-official repos, scan them and check for image vulnerabilities before using them for further application deployment. Scanning tools like Clair, ThreatMapper, and Trivy are excellent free open source tools for scanning image vulnerabilities.

DevOps engineers nowadays use Dockerfile extensively for application deployment. When using Dockerfiles, follow these security practices:

- Add non-root user to Docker image
 - Add health check
 - Remove unwanted packages from the image
 - Use multi-stage build
 - Use COPY instead of ADD
 - Use the official Docker image
- While running containers, follow these security best practices:
- Avoid using system reserved port numbers
 - Assign CPU/memory limit to the container

API calls

More than 85% of vulnerabilities are invoked during API calls. Docker has a Docker client installation feature which uses API calls. When dealing with such API calls we must take the necessary precautions. Using tools like nmap to scan those API-exposed calls is a good practice. **END** 

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